MIL-R-23481B(OS)
13 March 1967
SUPERSEDING
MIL-R-23481A(Wp)
1 March 1966

MILITARY SPECIFICATION

ROCKET MOTOR, MARK 30 MODS 1 AND 2

This specification has been approved by the
Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification covers rocket motor Mark 30 Mods 1 and 2.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibrations of Shipboard Equipment
MIL-STD-414	Sampling Procedures and Tables for Inspection by Variables for Percent Defective

DRAWINGS

Bureau of Naval Weapons

LM 2177145	Loaded Assembly, Rocket Motor, Mark 30 Mod 1
2177155	Ballistic Requirements, Rocket Motor, Mark 30 Mods 1 and 2

FSC 1336

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LM 2177157 Loaded Assembly, Rocket Motor, Mark 30 Mod 2

(When requesting any of the preceding documents, give the complete title and designation. Copies of this specification and other unclassified specifications and standards required by contractors in connection with specific procurement functions should be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120. Drawings and Ordnance Data (OD) should be obtained from the Commanding Officer, Naval Ordnance Plant (CTDO), Iroquois Station, Louisville, Kentucky, 40214).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

PUBLICATIONS

General Services Administration

Code of Federal Regulations

Title 49 Transportation

(The Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles are a part of the Code of Federal Regulations. Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402. Orders should cite 49 CFR 71-90, current revision.)

3. REQUIREMENTS

3.1 Preproduction sample. Unless otherwise specified in the contract or order (see 6.2), a preproduction sample of at least 15 rocket motors shall be manufactured using the materials and procedures proposed for production. The preproduction sample shall be inspected as specified in section 4 for the purpose of determining that, prior to starting production, the supplier's methods are capable of providing rocket motors that conform to the technical requirements specified herein. After satisfactorily passing all preproduction inspections, no changes in materials or procedures shall be made without prior written approval of the procuring activity.

3.2 Data. No data is required by this specification, or by applicable documents referenced in section 2, unless specified in the contract or order (see 6.2).

3.3 Description. The rocket motor shall be in accordance with Drawing LM 2177145 or LM 2177157 and as specified herein.

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3.4 Ballistic performance. The rocket motor ballistic performance shall be in accordance with Drawing 2177155.

3.5 Preproduction environmental testing. The rocket motor shall perform as specified herein after being subjected to the environmental tests of 4.7.2.2.

3.6 Workmanship. The rocket motors shall be uniform in quality and free of defects that could adversely affect use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection lot. An inspection lot of motors shall consist of all motors produced under homogenous manufacturing conditions and presented for acceptance at one time.

4.3 Classification of inspections. Inspections shall be classified as follows:

(a) Preproduction inspection (see 4.5)
(b) Production quality conformance inspection (see 4.6)
(c) Preparation for delivery inspection (see 4.8).

4.4 Sampling.

4.4.1 General inspection. Unless otherwise specified herein, sampling for determination of acceptance of rocket motors shall be in accordance with MIL-STD-105, inspection level II, double sampling plans for normal inspection.

4.4.2 Sampling for lot acceptance based on performance. Sampling for lot acceptance based on performance shall be in accordance with table I.

Table I. Sampling for Lot Acceptance Based on Performance

Motor Lot Size	Ballistic Performance Sample Size	AQL	Other than Ballistic Performance Sample Size	AQL
1 through 12	1	--	1	--
13 through 25	3	--	3	--
26 and over	MIL-STD-414	0.25	MIL-STD-105*	1.0

* Inspection level II, single sampling plan for normal inspection.

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4.5 Preproduction inspection. A minimum of 15 preproduction motors, manufactured in accordance with 3.1, shall be submitted to an activity designated by the procuring activity for inspection to determine conformance to the requirements of this specification. Inspection shall be within 90 days after submission of the sample and be performed with equipment and procedures specified by the procuring activity. Inspection shall consist of all the examinations and tests of 4.7 and 4.8.

4.6 Production quality conformance inspection.

4.6.1 Examination. Each motor shall be examined as specified in 4.7.1 and 4.8.1.

4.6.2 Tests. The quality conformance motors selected in accordance with 4.4 shall be tested as specified in 4.7.2.1 and 4.7.2.3.

4.7 Inspection procedure.

4.7.1 Examination. Visually examine each motor for conformance to 3.6. Presence of one or more defects shall be cause for rejection.

4.7.2 Tests.

4.7.2.1 General acceptance inspection. The rocket motors shall be inspected for conformance to the requirements of 3.3. Acceptance shall be based on the sampling plans of 4.4.1 and the Acceptable Quality Levels (AQL) of table II.

Table II. Acceptable Quality Levels

Classification of Characteristics	AQL
Critical	*
Major	1.0 percent
Minor	2.5 percent

*For inspection lot sizes of 125 motors or less, inspect 100 percent of the lot; for lot sizes of 126 to 3200, use a sample of 125 motors with acceptance number zero; for lot sizes greater than 3200, sample in accordance with MIL-STD-105 with acceptance number zero.

4.7.2.2 Environmental. The preproduction lot of rocket motors shall be subjected to environmental tests prior to static firing in accordance with the schedule of table III.

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Table III. Environmental Test and Static Firing Temperature Requirements

Motor Number	Temperature Cycling	Vibration	Shock	Longitudinal Acceleration	Lateral Acceleration	Firing Temperature, °F
1	X					120
2	X					10
3		X				10
4			X			10
5			X			120
6				X		120
7					X	120
8	X	X				10
9	X	X				120
10	X		X			10
11	X		X			120
12		X	X			10
13		X	X			120
14	X	X	X	X	X	10
15	X	X	X			120

4.7.2.2.1 Temperature cycling. The rocket motors shall be subjected to temperature cycling. The soak time at each temperature shall be a minimum of 48 hours. Temperature tolerances for environmental tests and static firings shall be ± 5° Fahrenheit (F). The test is composed of three cycles with a single cycle described as shown in table IV.

Table IV. Temperature Conditioning Cycle

Step	Temperature Gradient
1	77° to 120°F
2	120° to 77°F
3	77° to 0°F
4	0° to 77°F

4.7.2.2.2 Vibration.

(a) The round (see 6.6) shall be mounted on the launcher and undergo the vibrations of table V distributed about a 1 gravity (g) steady state loading at a 0° launcher elevation. Tolerance values for vibration and shock tests shall be: frequency ± 2 percent, acceleration ± 10 percent, and g ± 5 percent.

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Table V. Vibration Testing

Launcher Condition	Frequency*, cycles per second (cps)	Acceleration, g	Minimum Duration, minutes
Synchronizing	7	3.2	10
Tracking	7	3.2	50
Stationary	In accordance with MIL-STD-167, type I.		

*Simple harmonic motion

(b) A linear vibration with high frequency shock transients superimposed shall be applied in the transverse direction of both control surface planes at the tail of the round. The linear vibration shall be swept from 20 to 600 cps and shall produce a 300-pound peak force over the entire frequency range. The high frequency shock transients shall have a repetition rate of not less than 100 cps and shall produce 30 g peak transients simultaneously along the longitudinal axis of the round as well as in the control surface planes.

(c) A torsional vibration shall be applied about the longitudinal axis at the after end of the round in an exponential frequency sweep from 20 to 300 cps capable of producing a 3,000 inch-pound torque over the entire frequency range.

4.7.2.2.3 Shock. The round shall be subjected to the following shock tests:

(a) Longitudinal shock rising to 40 g in 0.010 ± 0.003 second with a minimum dwell time of 0.020 second
(b) Torsional shock of 1600 radians/second^2.

4.7.2.2.4 Acceleration. The rocket motor shall be subjected to the following acceleration tests:

(a) Longitudinal acceleration of 40 g held for a minimum of 4 seconds
(b) Lateral acceleration of 30 g held for a minimum of 4 seconds.

4.7.2.3 Performance. Rocket motors, selected in accordance with 4.4, shall be statically fired with equipment and procedures approved by the procuring activity. The preproduction rocket motors shall be fired at the temperatures specified in table III. The first quality conformance rocket motor shall be statically fired at a temperature of $10^{\circ} \pm 5^{\circ}$F and each succeeding motor shall be alternately fired at $120^{\circ} \pm 5^{\circ}$F and $10^{\circ} \pm 5^{\circ}$F. Rocket motors shall be conditioned to the static firing temperature for a minimum of 48 hours. The motors shall be fired within 1 hour after removal from conditioning. If a motor is not fired within 1 hour, the motor shall be returned to conditioning until equilibrium temperature is reached. Four records, two thrust and two pressure, shall be obtained for each firing. Two complete channels of instrumentation shall be used for

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both pressure and thrust. Ballistic data shall be reported as the average of two channels unless one channel is obviously in error.

4.7.2.3.1 Performance lot acceptance. Failure of a lot acceptance firing to meet ballistic requirements due to propellant shall be cause for rejection of all grains manufactured from the same batch of propellant. The remaining motors shall be re-grouped for lot acceptance tests. Failure of a lot acceptance firing for reasons other than ballistics shall be cause for rejection of the lot. When three motors are fired, acceptance shall be based on all three motors meeting the required performance. When more than three motors are fired, the sampling plans and acceptance provisions of table I shall be applied for determining whether the requirements of this specification are met.

4.8 Preparation for delivery inspection.

4.8.1 Examination. Each rocket motor shall be examined for conformance to section 5. Presence of one or more defects shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. Unless otherwise specified by the procuring activity (see 6.2), packaging and packing shall be in accordance with MIL-P-116, method III.

5.2 Marking. Each rocket motor container shall be marked in accordance with 49 CFR 71-90 and MIL-STD-129. Marking shall include, but not be limited to,the following information:

(a) Title (with applicable mod number), number, and revision letter of this specification

(b) Supplier's name and location

(c) Lot number, batch number, and date of manufacture

(d) Net weight and rocket motor serial number

(e) Storage limitations.

6. NOTES

6.1 Intended use. The rocket motor covered by this specification is intended for use as a missile sustainer.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following:

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(a) Title, number, and revision letter of this specification
(b) When a preproduction sample is not required (see 3.1 and 6.3)
(c) Preproduction test activity (see 4.5 and 6.4)
(d) Minimum lot size, if applicable
(e) Data to be retained or submitted (see 3.2 and 6.5)
(f) Packaging and packing instructions (see 5.1)
(g) Place of delivery.

6.3 Preproduction sample. A preproduction sample is required from each supplier who has not produced the Mk 30 Mod 1 or Mod 2 rocket motors. If rocket motor production has been interrupted for more than 6 months or changes are requested in production methods, a preproduction sample may be required.

6.4 Preproduction test activity. The contracting officer should specify the name and location of the activity responsible for preproduction testing. Motor delivery schedules, location and source of test equipment, and test procedures should be specified.

6.5 Ballistic test data. The contracting officer should specify that computations and data for ballistic evaluation be reported on NAVWEP Form 8821, that data be traceable to rocket motor, and that static firing traces be reproduced and submitted as required.

6.6 Round. The term "round" means an assembled missile-booster combination. An inert loaded booster or simulated weights for missile components other than the rocket motor may be used during testing.

6.7 Explosives safety precautions.

6.7.1 Applicability. The loading, assembly, and handling of the explosives, subassemblies, and the finished items covered by this specification involve hazardous operations and therefore require suitable explosives safety precautions. At plants of the Naval Ordnance Systems Command, Department of the Navy, and other Government plants where the Government is responsible for safety, the precautions listed below are mandatory. These same minimum provisions listed below are recommended for observance by a contractor or manufacturer that explosively loads the item covered in this specification in order to fulfill his own safety responsibility. Use of this specification should not be construed to replace other safety regulations of governmental agencies or to relieve the contractor or manufacturer of responsibility for the safety of his operations.

6.7.2 Personnel and equipment. Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions and to limit damage to equipment and loading areas. In connection with the above, the precautionary measures in the following paragraphs should be observed. (See also 6.7.4 and other applicable parts of these safety precautions.)

6.7.2.1 Explosives operations should be kept under strict supervision. Periodic

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inspections should be employed to assure compliance with established safety regulations.

6.7.2.2 Safe equipment and methods should be utilized for handling and transporting explosives and loaded parts.

6.7.2.3 Where required, remote control barricaded handling equipment should be used for explosives operations such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.7.2.4 Employ properly proportioned and properly located protective barricades, screens, or shields at all required points.

6.7.2.5 Maintain equipment such as presses, dies, crimpers, blenders, mixers, sifters, conveyors, and the like, in safe operating condition.

6.7.2.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning, and associated industrial hazards.

6.7.2.7 During static tests, propulsion units such as rocket motors should be restrained against full thrust to prevent flight if ignited, for example, during test of igniter circuit. Also, means to vent flames and exhaust gases harmlessly should be provided.

6.7.3 Operation procedures. All loading operations should be conducted in full accordance with a formalized operating procedure which should include the following:

6.7.3.1 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operations.

6.7.3.2 Allow no fires or exposed electrical or other sparking equipment and little or no flammable material to be present in loading, handling, and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.7.3.3 Appropriate safety instructions for specific operations should be posted and personnel instructed as to the hazards present and precautions involved.

6.7.3.4 The number of working personnel and transients permitted in each work space should be established and posted in such areas. The numbers should be the minimum number required to accomplish necessary operations.

6.7.3.5 All loading operations should be conducted in a neat and orderly manner.

6.7.3.6 Enforce good housekeeping and maintain effective policing, inspection, and supervisory methods throughout the loading area and surroundings.

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6.7.3.7 Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from, floors, walls, ceiling, ledges, tables, benches, piping, and equipment or the items loaded. Clean up any spilled material immediately.

6.7.3.8 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.7.3.9 Personnel handling detonators, primers, delay elements, leadins, boosters, and related parts which affect functioning should, insofar as practicable, avoid using bare fingers or improper equipment, in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.7.3.10 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges (see 6.7.4).

6.7.4 Static and electrical.

6.7.4.1 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, where necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics, such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low energy type electric primers, detonators, and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare noninsulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding from induced electric currents originated by sources such as lightning, static, electromagnetic radiations, etc. Where necessary for safety, humidity of work rooms should be appropriately increased, as required, to lessen electrostatic effects but without inducing excessive moisture absorption by any components of the item being processed.

6.7.4.2 Protect all explosive operations from effects of electric current originating in equipment such as soldering irons or guns, heaters, switches, wiring, motors, lights,

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test instruments, etc, by suitable insulation, grounding, separation, or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or due to completing an electric circuit through an electric primer, detonator, or squib. For example, circuits may be inadvertently completed from a defective electric soldering iron through a grounded contact. All electric type primers, detonators, or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Removable short circuiting clips or other devices should be employed during manufacturing operations involving items such as electric primers, detonators, or squibs.

6.7.4.3 Where necessary, protect personnel and operations from electromagnetic energy. In certain work with very sensitive explosives or items, a shielded environment free from significant electromagnetic energy should be provided.

6.7.5 Operating facility and storage regulations.

6.7.5.1 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance, at a minimum, with the American Table of Distances (ATD); and, while in process, in safety lockers and chests if in loading rooms, or in adequate ready or service magazines located in accordance with Intraplant Distances set forth in the ATD when outside of loading rooms. The Armed Services Explosives Safety Board standards apply to Navy-managed or contractor-managed, Navy-owned explosives loading plants. These are standards approved by the Department of Defense, the Secretary of the Navy, and the Naval Ordnance Systems Command.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-R-23481B(OS) ROCKET MOTOR, MARK 30 MODS 1 and 2

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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